

DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 5 (AT2)

Q1

No.	Best Chart Type	Short Justification
1	Amounts – Bar Chart	A bar chart is best for comparing the population of different countries. The bar height makes the difference in population easy to see.
2	Distribution – Box Plot	A box plot shows how delivery times are spread for each courier. It also shows the median and unusual values.
3	Proportion – Pie Chart	A pie chart shows how the total budget is divided. Each slice represents the share of rent, food, transport, and savings.
4	x-y Relationship – Scatter Plot	A scatter plot shows the relationship between study hours and marks. It helps identify whether more study time is linked with higher marks.
5	Temporal – Line Chart	A line chart is suitable because the data changes month by month. Three lines can compare the temperature trends of the three cities.
6	Network – Network Graph	A network graph shows airports as nodes and flight routes as links. It clearly shows which airports are directly connected.

Q2

Code:

```
library(ggplot2)

# Create the data
app_data <- data.frame(
  App_Category = c("Social Media", "Gaming", "Productivity",
    "Finance", "Health & Fitness", "Education"),
  Downloads_Millions = c(58, 46, 27, 19, 33, 24)
)

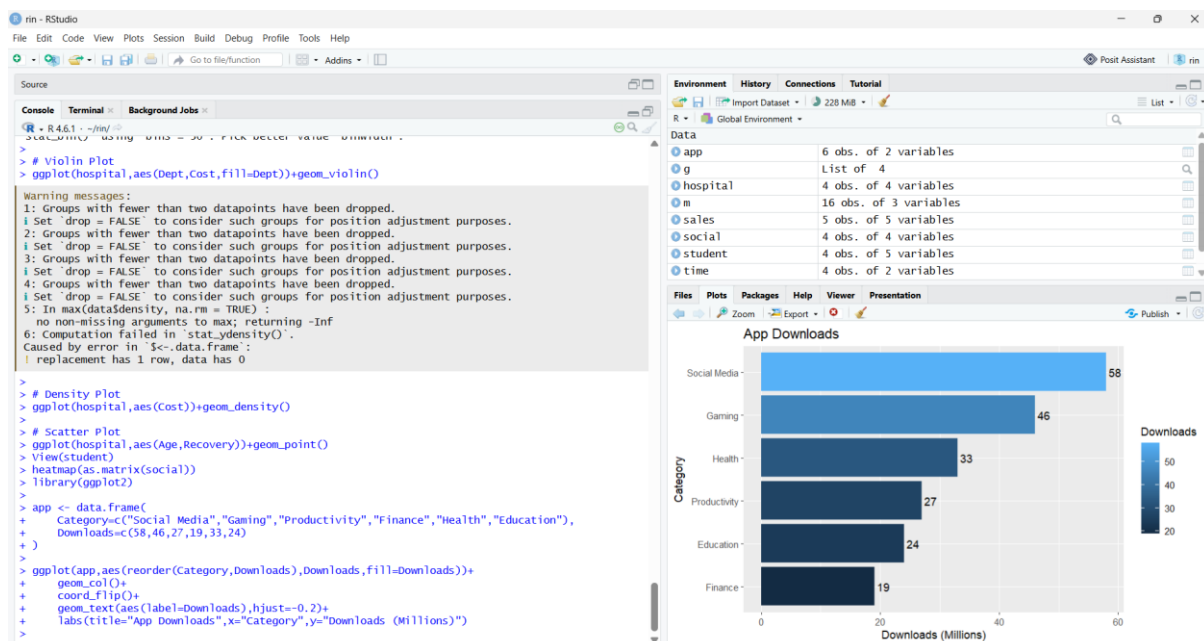
# Sort data in ascending order
app_data <- app_data[order(app_data$Downloads_Millions), ]

# Create horizontal bar plot
ggplot(app_data, aes(
```

```

x = Downloads_Millions,
y = reorder(App_Category, Downloads_Millions),
fill = Downloads_Millions
)) +
geom_col() +
geom_text(
  aes(label = Downloads_Millions),
  hjust = -0.2
) +
scale_fill_gradient(low = "lightblue", high = "darkblue") +
labs(
  title = "App Downloads by Category",
  x = "Downloads (Millions)",
  y = "App Category"
) +
theme_minimal()

```



Q3

Code:

```

library(ggplot2)

traffic <- data.frame(
  Device=rep(c("Desktop","Mobile","Tablet"), each=4),
  Month=rep(c("Jan","Feb","Mar","Apr"), 3),
  Visits=c(45,48,42,50,80,88,95,102,20,18,17,15)
)

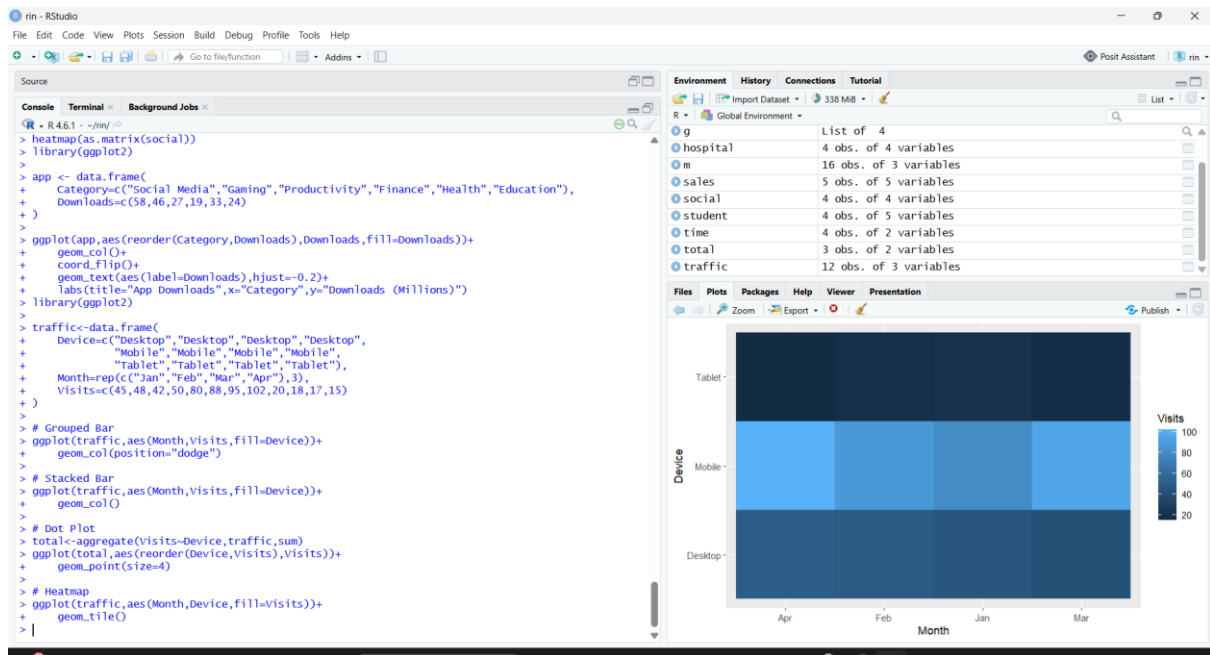
# Grouped Bar
ggplot(traffic,aes(Month,Visits,fill=Device))+
  geom_col(position="dodge")

# Stacked Bar
ggplot(traffic,aes(Month,Visits,fill=Device))+
  geom_col()

# Dot Plot
total <- aggregate(Visits~Device,traffic,sum)
ggplot(total,aes(Visits,reorder(Device,Visits)))+
  geom_point(size=4)

# Heatmap
ggplot(traffic,aes(Month,Device,fill=Visits))+
  geom_tile()+
  geom_text(aes(label=Visits))

```



Q4

Code:

```
library(ggplot2)
```

```
library(ggrridges)
```

```
set.seed(7)
```

```
delivery <- data.frame(
```

```
  Route = rep(c("A", "B", "C", "D"), each = 50),
```

```
  Time = c(
```

```
    rnorm(50, 32, 6),
```

```
    rnorm(50, 28, 4),
```

```
    rnorm(50, 40, 9),
```

```
    rnorm(50, 35, 5)
```

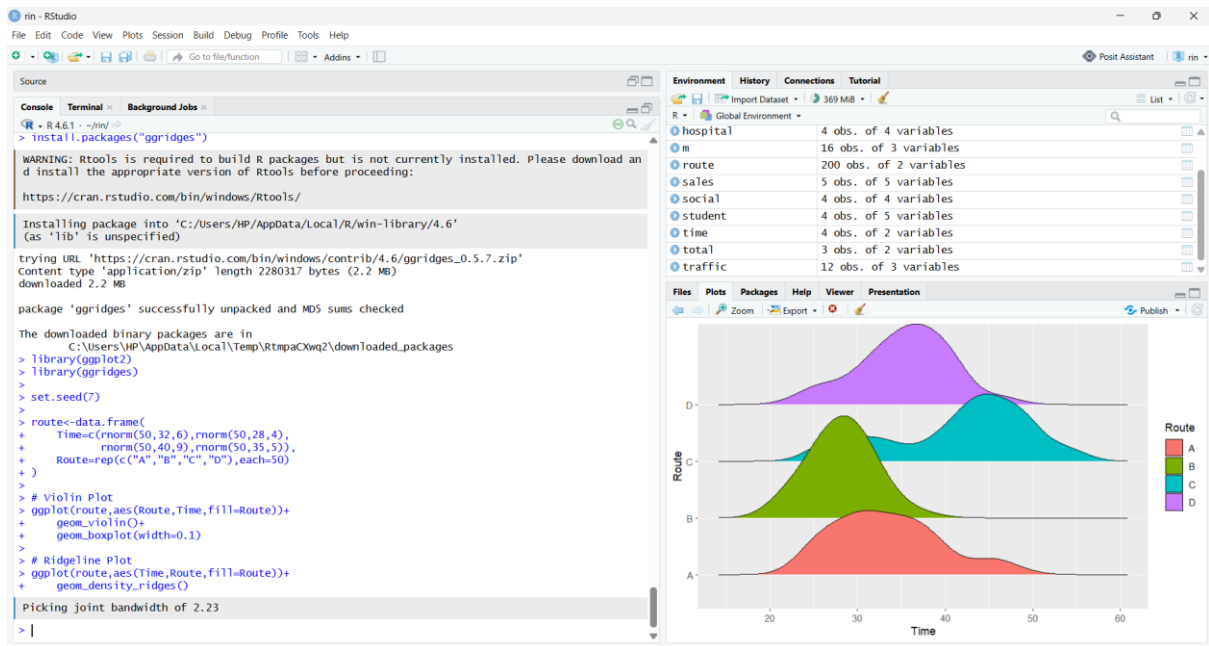
```
  )
```

```
)
```

```
# Violin Plot
```

```
ggplot(delivery, aes(Route, Time, fill = Route)) +
```

```
geom_violin(trim = FALSE) +  
geom_boxplot(width = 0.12, fill = "white") +  
labs(  
  title = "Delivery Time by Route",  
  x = "Route",  
  y = "Delivery Time (minutes)"  
) +  
theme_minimal()  
  
# Ridgeline Plot  
delivery$Route <- factor(  
  delivery$Route,  
  levels = c("B", "A", "D", "C")  
)  
  
ggplot(delivery, aes(Time, Route, fill = Route)) +  
  geom_density_ridges(alpha = 0.7) +  
  labs(  
    title = "Delivery Time Distribution by Route",  
    x = "Delivery Time (minutes)",  
    y = "Route"  
  ) +  
  theme_ridges()
```



Q5

Code:

```
library(ggplot2)
```

```
# Built-in Iris dataset
```

```
data(iris)
```

```
# 1. Histogram with KDE
```

```
ggplot(iris, aes(Petal.Length)) +  
  geom_histogram(aes(y = after_stat(density)), bins = 20) +  
  geom_density() +  
  labs(  
    title = "Petal Length Distribution",  
    x = "Petal Length",  
    y = "Density"  
  ) +  
  theme_minimal()
```

```
# 2. Histogram + KDE by Species
```

```
ggplot(iris, aes(Petal.Length, fill = Species)) +  
  geom_histogram(
```

```

aes(y = after_stat(density)),
bins = 20,
alpha = 0.4,
position = "identity"
) +
geom_density(alpha = 0.7) +
labs(
  title = "Petal Length Distribution by Species",
  x = "Petal Length",
  y = "Density"
) +
theme_minimal()

```

